Supplementary Material for:

Folded Transport MCMC: Eliminating Label Switching by Sampling on a Fundamental Domain

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This supplement accompanies the main paper "Folded Transport MCMC: Eliminating Label Switching by Sampling on a Fundamental Domain." It provides a self-contained certification toolkit, complete proofs, boundary analysis, and experimental details.

A Self-contained certification toolkit

This appendix restates the key results from the LCNF [Hu, 2026b] and CerT-MCMC [Hu, 2026a] frameworks that FolT-MCMC builds upon, so that the present paper can be read and verified independently.

A.1 Independence MH oscillation bound

For an independence Metropolis–Hastings chain with target π and proposal q, define the log-density ratio $h(x) = \log \pi(x) - \log q(x)$ and its oscillation on a set S as $\mathrm{osc}_S(h) = \sup_S h - \inf_S h$. The Mengersen–Tweedie bound [Mengersen and Tweedie, 1996] gives

$$\gamma \;\geq\; \frac{2}{1 + \exp(\mathrm{osc}(h))},$$

where the oscillation is over the full support. When only the HPD-core oscillation $\mathrm{osc}_{\mathcal{K}_\alpha}(h)$ is controlled, we define the *HPD-core certified lower-bound diagnostic*

$$\underline{\gamma}_\alpha \;:=\; \frac{2}{1+\exp(\mathrm{osc}_{\mathcal{K}_\alpha}(h))}.$$

This is a computable proxy for the spectral gap; it becomes a true spectral-gap lower bound only if the same oscillation bound holds on the full support, which requires additional tail control not provided by the present framework.

A.2 LCNF empirical oscillation theorem

The following is a simplified restatement of Hu [2026b, Theorem 5.1].

Theorem A.1 (LCNF empirical oscillation; restated)**.** *Let $\mathcal{K}_\alpha$ be the $(1-\alpha)$-HPD set of π with diameter $D_K = \mathrm{diam}(\mathcal{K}_\alpha)$ and density floor $\pi_{\min} = \inf_{\mathcal{K}_\alpha} \pi > 0$. Suppose h is continuously differentiable on an open neighbourhood of $\mathcal{K}_\alpha$ with gradient supremum $M_K = \sup_{\mathcal{K}_\alpha} \|\nabla h\|$. Draw $Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} \pi$ and define $\widehat{\mathrm{osc}}_n = \max_i h(Z_i) - \min_i h(Z_i)$. Let ε^* be the smallest $\varepsilon > 0$ satisfying the covering condition*

$$\Big(\frac{D_K}{\varepsilon}+1\Big)^d \big(1-\pi_{\min}\, V_d\, \varepsilon^d\big)^{n_{\text{eff}}} \;\leq\; \frac{\delta}{3}, \tag{1}$$

where V_d is the volume of the unit d-ball and $n_{\text{eff}} = n(1-\alpha) - \sqrt{n \cdot \frac{1}{2}\ln(3/\delta)}$. Then with probability $\geq 1-\delta$:

$$\mathrm{osc}_{\mathcal{K}_\alpha}(h) \;\leq\; \widehat{\mathrm{osc}}_n + 2\, M_K\, \varepsilon^*.$$

The key ingredients are: (i) a probabilistic covering argument showing that n i.i.d. samples from π form an ε^*-net of $\mathcal{K}_\alpha$ with high probability; (ii) a Lipschitz interpolation step using M_K to bound the oscillation at uncovered points; (iii) a Hoeffding-type concentration to convert the random sample count in $\mathcal{K}_\alpha$ to the effective count n_{eff}.

A.3 Quantile-core certificates

The CerT-MCMC extension [Hu, 2026a] addresses the sensitivity of the HPD-core bound to extreme density-ratio values. Given certification samples $\{h(Z_i)\}_{i=1}^n$, the ρ-trimmed quantile core excludes the fraction ρ of samples with the largest $|h(Z_i) - \mathrm{median}(h)|$ and computes the oscillation on the remaining $(1-\rho)n$ samples:

$$\widehat{\mathrm{osc}}_n^{\mathrm{QC}} = \max_{i \in \mathcal{I}_\rho} h(Z_i) - \min_{i \in \mathcal{I}_\rho} h(Z_i),$$

where $\mathcal{I}_\rho$ is the index set after trimming. The quantile-core certified lower-bound diagnostic is

$$\mathrm{QC}\,\underline{\gamma} := \frac{2}{1 + \exp(\widehat{\mathrm{osc}}_n^{\mathrm{QC}} + 2\,\widehat{M}_n^{\mathrm{QC}}\,\varepsilon^*)},$$

where $\widehat{M}_n^{\mathrm{QC}}$ is the gradient supremum over the trimmed samples. This diagnostic is more robust than the untrimmed version but further from a full-support guarantee. All experiments in the present paper report QC $\underline{\gamma}$ at $\rho = 0.05$.

A.4 From LCNF to quotient space: what FolT-MCMC adds

FolT-MCMC transfers Theorem A.1 to the quotient setting by making three substitutions:

1. $\mathcal{K}_\alpha \to \mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D$ (HPD identity, Proposition 3.2);
2. $\pi_{\min} \to \pi_{\min}^F \geq s\,\pi_{\min}$ (density-floor improvement, Corollary 3.3);
3. Euclidean metric $\to$ quotient metric $d_G(x, y) = \min_{g \in G} \|x - g \cdot y\|$, with the Lipschitz property of h_F established in Lemma 3.8 (local version, $d_G(x, y) \leq \varepsilon_F^*$).

The covering condition (1) carries over with $D_K \to D_F$, $\pi_{\min} \to \pi_{\min}^F$, and n_{eff} unchanged, yielding the folded certificate (Theorem 4.1). The proofs of all FolT-MCMC theorems are given in Sections B.1–B.7 below.

B Complete proofs

B.1 Proof of Proposition 3.2 (HPD identity)

Proof. Let $U(\theta) = -\log \pi(\theta) + \log Z_\pi$ denote the potential of π, where Z_π is the normalising constant. The folded potential is $U_F(z) = -\log \pi_F(z) + \log Z_{\pi_F} = -\log(s\pi(z)) + \log 1 = U(z) - \log s$, using $Z_{\pi_F} = 1$ since π_F integrates to one on D.

Part (i). We show that $U(Z_F)$ under $Z_F \sim \pi_F$ has the same distribution as $U(\Theta)$ under $\Theta \sim \pi$. For any Borel function $\varphi \colon \mathbb{R} \to [0, \infty)$, the G-invariance of π and fundamental-domain decomposition give

$$\begin{aligned}
\mathbb{E}_{\Theta\sim\pi}[\varphi(U(\Theta))] &= \int_{\mathbb{R}^d} \varphi(U(\theta))\,\pi(\theta)\,\mathrm{d}\theta \\
&= \sum_{g\in G} \int_{gD} \varphi(U(\theta))\,\pi(\theta)\,\mathrm{d}\theta \\
&= \sum_{g\in G} \int_{D} \varphi(U(g\cdot z))\,\pi(g\cdot z)\,\mathrm{d}z \qquad \text{(substituting } \theta = g\cdot z,\ |\det g| = 1) \\
&= s\int_{D} \varphi(U(z))\,\pi(z)\,\mathrm{d}z \qquad \text{(since } U(g\cdot z) = U(z) \text{ and } \pi(g\cdot z) = \pi(z)) \\
&= \int_{D} \varphi(U(z))\,\pi_F(z)\,\mathrm{d}z = \mathbb{E}_{Z_F\sim\pi_F}[\varphi(U(Z_F))]. \qquad (2)
\end{aligned}$$

Since (2) holds for all non-negative Borel φ, the distributions of $U(\Theta)$ and $U(Z_F)$ coincide. In particular, their $(1-\alpha)$-quantiles are equal: the $(1-\alpha)$-quantile of $U(Z_F)$ is u_α. Since $U_F(Z_F) = U(Z_F) - \log s$, the $(1-\alpha)$-quantile of $U_F(Z_F)$ is $u_\alpha - \log s =: u_\alpha^F$.

Part (ii).

$$\begin{aligned}
\mathcal{K}_\alpha^F &= \{z \in D : U_F(z) \le u_\alpha^F\} = \{z \in D : U(z) - \log s \le u_\alpha - \log s\} \\
&= \{z \in D : U(z) \le u_\alpha\} = \mathcal{K}_\alpha \cap D.
\end{aligned}$$

□

B.2 Proof of Corollary 3.3

Proof. **Part (a).** Since $\mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D \subseteq \mathcal{K}_\alpha$, we have $\mathrm{diam}(\mathcal{K}_\alpha^F) \le \mathrm{diam}(\mathcal{K}_\alpha)$.

Part (b).

$$\pi^F_{\min} = \inf_{z \in \mathcal{K}^F_\alpha} \pi_F(z) = s \cdot \inf_{z \in \mathcal{K}_\alpha \cap D} \pi(z).$$

Since $\mathcal{K}_\alpha \cap D \subseteq \mathcal{K}_\alpha$, $\inf_{\mathcal{K}_\alpha \cap D} \pi \geq \inf_{\mathcal{K}_\alpha} \pi = \pi_{\min}$, so $\pi^F_{\min} \geq s\, \pi_{\min}$.

When π is G-invariant and all orbits in $\mathcal{K}_\alpha$ have a representative in D (which holds by definition of a fundamental domain), the G-invariance implies that $\operatorname{ess\,inf}_{\mathcal{K}_\alpha \cap D} \pi = \operatorname{ess\,inf}_{\mathcal{K}_\alpha} \pi$, giving $\pi^F_{\min} = s\, \pi_{\min}$ in the essential-infimum sense. □

B.3 Proof of Lemma 3.8 (quotient Lipschitz constant)

Proof. Both π and $\theta \mapsto \sum_g q_{T_F}(g \cdot \theta)$ are G-invariant, so $h_F(g \cdot \theta) = h_F(\theta)$ for all $g \in G$ and $\theta \in \mathbb{R}^d$.

Fix $x, y \in \mathcal{K}^F_\alpha$ with $d_G(x, y) \leq \varepsilon^*_F$. Let $g^* = \arg\min_{g \in G} \|x - g \cdot y\|$, so that $\|x - g^* \cdot y\| = d_G(x, y) \leq \varepsilon^*_F$. The segment from x to $g^* \cdot y$ lies entirely within the Euclidean ε^*_F-ball centred at x. Since $x \in \mathcal{K}^F_\alpha$, this ball is contained in $\{w \in \mathbb{R}^d : d_G(w, \mathcal{K}^F_\alpha) \leq \varepsilon^*_F\}$. By the mean value theorem applied along this segment,

$$|h_F(x) - h_F(g^* \cdot y)| \;\leq\; \sup_{w:\, d_G(w, \mathcal{K}^F_\alpha) \leq \varepsilon^*_F} \|\nabla h_F(w)\| \;\cdot\; \|x - g^* \cdot y\| \;=\; M_F \cdot d_G(x, y).$$

Using $h_F(g^* \cdot y) = h_F(y)$ (G-invariance) completes the proof. □

B.4 Proof of Theorem 4.1 (FolT-MCMC convergence diagnostic)

Proof. The proof instantiates Hu [2026b, Theorem 5.1] on the quotient setting. We verify that the three inputs required by that theorem are available.

Input 1: Covering number bound. Since $d_G(x, y) \leq \|x - y\|$, any Euclidean ε-cover of $\mathcal{K}^F_\alpha$ is also a d_G-ε-cover. The standard volumetric argument gives

$$\mathcal{N}_G(\mathcal{K}^F_\alpha, \varepsilon) \leq \mathcal{N}_E(\mathcal{K}^F_\alpha, \varepsilon) \leq \left(\frac{2D_F}{\varepsilon} + 1\right)^d,$$

where $D_F = \operatorname{diam}(\mathcal{K}^F_\alpha)$.

Input 2: Ball-mass lower bound. Under Assumption 3.7, $|\mathrm{Stab}(z)| = 1$ for all $z \in \mathcal{K}_\alpha^F$. For such z, the quotient ball $B_G(z, \varepsilon) = \{z' : d_G(z, z') \le \varepsilon\}$ corresponds in $\mathbb{R}^d$ to the union $\bigcup_{g\in G}(B(g \cdot z, \varepsilon) \cap g \cdot D)$. The s pieces tile a full Euclidean ball (up to the measure-zero boundaries $\partial(g \cdot D)$), so $\mathrm{Vol}(B_G(z, \varepsilon)) = V_d\, \varepsilon^d$. Therefore

$$\pi_F(B_G(z, \varepsilon) \cap D) \;\ge\; \pi_{\min}^F \cdot V_d \cdot \varepsilon^d.$$

Input 3: Lipschitz interpolation. Lemma 3.8 applies to each point $x \in \mathcal{K}_\alpha^F$ and its nearest certification sample Z_i, since the covering event guarantees $d_G(x, Z_i) \le \varepsilon_F^*$.

Substituting these three inputs into Hu [2026b, Theorem 5.1], the covering condition

$$\Big(\frac{D_F}{\varepsilon} + 1\Big)^d \big(1 - \pi_{\min}^F\, V_d\, \varepsilon^d\big)^{n_{\mathrm{eff}}} \le \frac{\delta}{3}$$

guarantees that with probability $\ge 1 - \delta$, every point of $\mathcal{K}_\alpha^F$ lies within d_G-distance ε_F^* of some certification sample. The oscillation bound follows:

$$\mathrm{osc}_{\mathcal{K}_\alpha^F}(h_F) \le \widehat{\mathrm{osc}}_n^F + 2M_F\varepsilon_F^*.$$

Applying the Mengersen–Tweedie lower-bound functional to the certified HPD-core oscillation gives the diagnostic value $\underline{\gamma}_{F,\alpha} := 2/(1 + \exp(\widehat{\mathrm{osc}}_n^F + 2M_F\varepsilon_F^*))$. □

B.5 Proof of Theorem 4.3 (covering radius improvement)

Proof. Write the left-hand side of the covering condition as

$$\Phi(D_K, \pi_{\min}, \varepsilon) \;=\; \Big(\frac{D_K}{\varepsilon} + 1\Big)^d \big(1 - \pi_{\min}\, V_d\, \varepsilon^d\big)^{n_{\mathrm{eff}}}.$$

Monotonicity in D_K. For fixed ε and $\pi_{\min}$, Φ is increasing in D_K (the first factor $(D_K/\varepsilon + 1)^d$ increases).

Monotonicity in $\pi_{\min}$. For fixed ε and D_K, Φ is decreasing in $\pi_{\min}$ (the base $1 - \pi_{\min}\, V_d\, \varepsilon^d$ decreases, and since $n_{\mathrm{eff}} > 0$ and the base is in $(0, 1)$, the second factor decreases).

By Corollary 3.3: (a) $D_F = \mathrm{diam}(\mathcal{K}_\alpha^F) \leq D_U = \mathrm{diam}(\mathcal{K}_\alpha)$; (b) $\pi_{\min}^F \geq s\,\pi_{\min} \geq \pi_{\min}$. Therefore, for every $\varepsilon > 0$,

$$\Phi(D_F,\, \pi_{\min}^F,\, \varepsilon) \;\leq\; \Phi(D_U,\, \pi_{\min},\, \varepsilon).$$

Since ε_U^* satisfies $\Phi(D_U, \pi_{\min}, \varepsilon_U^*) \leq \delta/3$, we have $\Phi(D_F, \pi_{\min}^F, \varepsilon_U^*) \leq \delta/3$. By the minimality of ε_F^* (smallest ε satisfying the folded covering condition), $\varepsilon_F^* \leq \varepsilon_U^*$. □

B.6 Proof of Lemma 4.5 (oscillation gap)

Proof. Let $h_U = \log \pi - \log q_U$ (unfolded log-density ratio) and $h_F = \log \pi_F - \log q_F$ (folded log-density ratio). We bound $\widehat{\mathrm{osc}}_n$ from below and $\widehat{\mathrm{osc}}_n^F$ from above.

Step 1: Lower bound on $\mathrm{osc}_{\mathcal{K}_\alpha}(h_U)$. The HPD set decomposes into connected components $\mathcal{K}_\alpha = \bigcup_{j=1}^k \mathcal{K}_\alpha^{(j)}$ (one per mode). Condition (M′) gives, for each $j \neq 1$,

$$\inf_{\theta \in \mathcal{K}_\alpha^{(j)}} h_U(\theta) \;\geq\; \sup_{\theta \in \mathcal{K}_\alpha^{(1)}} h_U(\theta) + \Delta_j.$$

Therefore

$$\begin{aligned}
\mathrm{osc}_{\mathcal{K}_\alpha}(h_U) &\geq \sup_{\mathcal{K}_\alpha^{(j)}} h_U - \inf_{\mathcal{K}_\alpha^{(1)}} h_U \\
&\geq \inf_{\mathcal{K}_\alpha^{(j)}} h_U - \inf_{\mathcal{K}_\alpha^{(1)}} h_U \\
&\geq \Big(\sup_{\mathcal{K}_\alpha^{(1)}} h_U + \Delta_j\Big) - \inf_{\mathcal{K}_\alpha^{(1)}} h_U \;\geq\; \Delta_j.
\end{aligned}$$

Taking the minimum over $j \neq 1$: $\mathrm{osc}_{\mathcal{K}_\alpha}(h_U) \geq \min_{j \neq 1} \Delta_j$.

The empirical oscillation $\widehat{\mathrm{osc}}_n$ is computed from n i.i.d. samples from π. By standard extreme-value concentration (see, e.g., Hu 2026b, Lemma 5.3),

$$\mathbb{P}\big(\mathrm{osc}_{\mathcal{K}_\alpha}(h_U) - \widehat{\mathrm{osc}}_n > t\big) \leq 2\exp(-2nt^2/R^2)$$

for a range parameter R. Setting the right-hand side to $\delta'/2$ gives $\widehat{\operatorname{osc}}_n \ge \operatorname{osc}_{\mathcal{K}_\alpha}(h_U) - c'_n$ with probability $\ge 1-\delta'/2$, where $c'_n = O(\sqrt{\log(2/\delta')/n})$.

Step 2: Upper bound on $\operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F)$. For $z \in \mathcal{K}_\alpha^F = \mathcal{K}_\alpha^{(1)} \cap D$ (a.e. by Proposition 3.2),

$$h_F(z) = \log(s\,\pi(z)) - \log q_F(z) = \underbrace{\left[\log(s\,\pi(z)) - \log q_{T_F}(z)\right]}_{=:\tilde{h}(z)} - \underbrace{\log\!\left(1 + \frac{\sum_{g\neq e} q_{T_F}(g\cdot z)}{q_{T_F}(z)}\right)}_{=:\rho(z)}.$$

By condition (W), $\operatorname{osc}_{\mathcal{K}_\alpha^F}(\tilde{h}) \le 2r_1$.

By condition (R), $|\rho(z)| \le r_F$ for all $z \in \mathcal{K}_\alpha^F$, so $\operatorname{osc}_{\mathcal{K}_\alpha^F}(\rho) \le r_F$.

Since $h_F = \tilde{h} - \rho$,

$$\operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F) \le \operatorname{osc}_{\mathcal{K}_\alpha^F}(\tilde{h}) + \operatorname{osc}_{\mathcal{K}_\alpha^F}(\rho) \le 2r_1 + r_F.$$

By the same concentration argument, $\widehat{\operatorname{osc}}_n^F \le \operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F) + c''_n$ with probability $\ge 1-\delta'/2$, where $c''_n = O(\sqrt{\log(2/\delta')/n})$.

Step 3: Combining. By a union bound over the two concentration events (probability $\ge 1-\delta'$):

$$\begin{aligned}
\widehat{\operatorname{osc}}_n - \widehat{\operatorname{osc}}_n^F &\ge \left(\operatorname{osc}_{\mathcal{K}_\alpha}(h_U) - c'_n\right) - \left(\operatorname{osc}_{\mathcal{K}_\alpha^F}(h_F) + c''_n\right) \\
&\ge \min_{j\neq 1} \Delta_j - (2r_1 + r_F) - (c'_n + c''_n) \\
&= \min_{j\neq 1} \Delta_j - 2r_1 - r_F - c_n,
\end{aligned}$$

where $c_n = c'_n + c''_n = O(\sqrt{\log(2/\delta')/n})$. □

B.7 Proof of Theorem 4.6 (diagnostic improvement)

Proof. Write

$$\begin{aligned}
C_U - C_F &= \left(\widehat{\operatorname{osc}}_n + 2M_U\varepsilon_U^*\right) - \left(\widehat{\operatorname{osc}}_n^F + 2M_F\varepsilon_F^*\right) \\
&= \left(\widehat{\operatorname{osc}}_n - \widehat{\operatorname{osc}}_n^F\right) + 2\left(M_U\varepsilon_U^* - M_F\varepsilon_F^*\right).
\end{aligned}$$

By Lemma 4.5 (with probability $\geq 1-\delta'$), $\widehat{\operatorname{osc}}_n - \widehat{\operatorname{osc}}_n^F \geq \Delta := \min_j \Delta_j - 2r_1 - r_F - c_n$.

Case 1: $M_F\varepsilon_F^* \leq M_U\varepsilon_U^*$. Then $C_U - C_F \geq \Delta + 0 = \Delta > 0$ (by assumption $\Delta > 0$).

Case 2: $M_F\varepsilon_F^* > M_U\varepsilon_U^*$. The sufficient condition (17) gives $\Delta > 2(M_F\varepsilon_F^* - M_U\varepsilon_U^*)$, so

$$C_U - C_F \geq \Delta - 2(M_F\varepsilon_F^* - M_U\varepsilon_U^*) > 0.$$

In both cases, $C_F < C_U$. The function $C \mapsto 2/(1+e^C)$ is strictly decreasing, so

$$\underline{\gamma}_{F,\alpha} = \frac{2}{1+e^{C_F}} > \frac{2}{1+e^{C_U}} = \underline{\gamma}_{U,\alpha}.$$

The overall confidence is $\geq 1-\delta_U-\delta_F-\delta'$ by a union bound over the three events: folded certificate (prob $\geq 1-\delta_F$), unfolded certificate ($\geq 1-\delta_U$), oscillation-gap concentration ($\geq 1-\delta'$). $\square$

C Boundary analysis

C.1 Fold boundary density for Gaussian mixtures

For the symmetric two-component Gaussian mixture $\pi = \frac{1}{2}\mathcal{N}(\mu_1, \sigma^2 I_d) + \frac{1}{2}\mathcal{N}(\mu_2, \sigma^2 I_d)$ with $\mu_1 = (\delta/2, 0, \dots, 0)$, $\mu_2 = -\mu_1$, and reflection fold boundary $\partial D = \{\theta_1 = 0\}$:

The density on the fold boundary is

$$\pi(0, \theta_{2:d}) = \frac{1}{2}(2\pi\sigma^2)^{-d/2} \exp\Big(-\frac{\delta^2/4 + \|\theta_{2:d}\|^2}{2\sigma^2}\Big) \cdot 2 = (2\pi\sigma^2)^{-d/2} \exp\Big(-\frac{\delta^2/4 + \|\theta_{2:d}\|^2}{2\sigma^2}\Big).$$

The peak density (at μ_1) is

$$\pi(\mu_1) = \frac{1}{2}(2\pi\sigma^2)^{-d/2} + \text{negligible}.$$

The ratio at the midpoint $\theta = 0$ is

$$\frac{\pi(0)}{\pi(\mu_1)} \approx 2\exp\Big(-\frac{\delta^2}{8\sigma^2}\Big).$$

For $\delta = 6\sigma$: ratio $\approx 2\exp(-4.5) \approx 0.022$.

This confirms that the fold boundary lies in a low-density valley, with density approximately 2% of the peak.

C.2 Condition for $\mathcal{K}_\alpha^F \cap \partial D = \varnothing$

The HPD set $\mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D$ does not touch the fold boundary if and only if $\mathcal{K}_\alpha \cap \partial D = \varnothing$. For the Gaussian mixture above, $\mathcal{K}_\alpha$ consists of two balls of radius $R_\alpha = \sigma\sqrt{\chi^2_{d,1-\alpha}}$ centred at $\pm\mu_1 = (\pm\delta/2, 0, \ldots, 0)$. The closest point of $\mathcal{K}_\alpha^{(1)}$ to ∂D has $\theta_1 = \delta/2 - R_\alpha$. The separation condition is

$$\mathcal{K}_\alpha^F \cap \partial D = \varnothing \quad \Longleftrightarrow \quad \delta/2 > R_\alpha \quad \Longleftrightarrow \quad \delta > 2\sigma\sqrt{\chi^2_{d,1-\alpha}}.$$

d	$\chi^2_{d,0.95}$	$2\sigma\sqrt{\chi^2}$	$\delta = 6\sigma$ OK?
2	5.99	4.90	Yes
5	11.07	6.65	No
10	18.31	8.56	No
20	31.41	11.21	No

For $\delta = 6\sigma$, the condition holds only at $d = 2$. At $d \geq 5$, the HPD set touches the fold boundary. However, the experiments in Section 5.3 show strong certified improvement at all dimensions, indicating that the boundary-touching case is handled adequately by the quotient-metric covering (Section 3.4).

C.3 Stabiliser analysis for permutation folding

For label-switching with $G = S_m$, the fundamental domain is the sorted chamber $D = \{(\theta^{(1)}, \ldots, \theta^{(m)}) : \theta_1^{(1)} \leq \cdots \leq \theta_1^{(m)}\}$.

A point $z \in D$ has non-trivial stabiliser if and only if two or more blocks share the same value of the sort coordinate: $\mathrm{Stab}(z) \supsetneq \{e\} \Longleftrightarrow \exists\, j \neq k : z_1^{(j)} = z_1^{(k)}$.

For well-separated components (centres $c_1 < c_2 < \cdots < c_m$ with $c_{j+1} - c_j \gg \sigma$), the HPD set $\mathcal{K}_\alpha^F$ is concentrated near the sorted centre $(c_1, \ldots, c_m)$, which has trivial stabiliser. The fixed-point stratum $\{z_1^{(j)} = z_1^{(k)}\}$ lies at distance $\geq (c_{j+1} - c_j)/2 - R_\alpha$ from $\mathcal{K}_\alpha^F$, which is positive for sufficiently separated components.

In this case Assumption 3.7 holds and $h_{\max} = 1$. When components are not well separated (as in the structural identification example of Section 6), $h_{\max}$ may exceed 1 and the covering condition uses the corrected mass factor $(s/h_{\max})\,\pi_{\min}^D\,V_d\,\varepsilon^d$.

D Experimental details

D.1 Architecture and hyperparameters

All flows use spectrally normalised RealNVP with tanh activations and scale clip $c = 0.7$. The architecture scales with dimension:

d	Layers L	Hidden dim h	Epochs	Batch size
≤ 4	8	64	2000	512
5–6	10	128	2000	512
8–12	12	128	3500	512
20	16	256	3500	512

Training uses Adam with learning rate 10^{-3} and annealed oscillation/gradient regularisation (FolT-OG-Anneal, as in Hu 2026a). The regularisation weight is annealed from 0 to its final value over the first 30% of training.

The folded and unfolded pipelines use identical architecture and hyperparameters for each d. Training samples are drawn from the target's exact sampler (for synthetic targets) or from a long-run random-walk Metropolis chain (for the structural identification application).

D.2 Certification protocol

Certification uses $n = 20\,000$ independent samples from the target (folded samples projected onto D for the folded pipeline). Quantile-core certificates are computed at $\rho \in \{0.025, 0.05, 0.10, 0.25\}$ with DKW-corrected confidence $\delta = 0.05$. The primary metric reported in all tables is the $\rho = 0.05$ certificate.

The gradient supremum M_F is estimated as the empirical maximum of $\|\nabla h_F(Z_i)\|$ over the certification samples, computed via automatic differentiation.

D.3 MCMC protocol

Independence Metropolis–Hastings with 4 chains of 5 000 steps each (10 000 for the RPS baseline comparison), burn-in equal to 20% of the chain length. Effective sample size (ESS) is estimated via batch means with batch size $\sqrt{n_{\text{chain}}}$.

For the structural identification application (Section 6), training samples are generated by a preliminary random-walk Metropolis run of 200 000 steps with adaptive step size, thinned to 50 000 samples.

D.4 Full quantile-core tables

Gaussian mixture, dimension scaling.

	QC $\underline{\gamma}$ at $\rho =$			
d	0.025	0.05	0.10	0.25
Unfolded				
2	0.216	0.402	0.608	0.810
5	0.031	0.094	0.260	0.565
10	0.005	0.016	0.065	0.290
20	0.005	0.016	0.062	0.280
FolT-MCMC				
2	0.890	0.936	0.968	0.990
5	0.895	0.941	0.970	0.991
10	0.870	0.922	0.958	0.986
20	0.850	0.902	0.950	0.983

Label switching.

		QC $\underline{\gamma}$ at $\rho =$			
Config	s	0.025	0.05	0.10	0.25
Unfolded					
m=2, p=2	2	0.142	0.366	0.571	0.790
m=3, p=2	6	0.019	0.066	0.185	0.486
m=3, p=4	6	0.010	0.036	0.111	0.380
m=4, p=2	24	0.001	0.0043	0.018	0.115
FolT-MCMC					
m=2, p=2	2	0.620	0.820	0.905	0.965
m=3, p=2	6	0.450	0.685	0.830	0.935
m=3, p=4	6	0.445	0.683	0.828	0.930
m=4, p=2	24	0.380	0.624	0.785	0.910

Banana diagnostic ($d = 2$, $s = 2$).

	QC $\underline{\gamma}$ at $\rho =$			
	0.025	0.05	0.10	0.25
Unfolded	0.064	0.272	0.537	0.797
FolT-MCMC	0.135	0.320	0.523	0.759

Note that at $\rho = 0.10$ and 0.25, the unfolded certificate slightly exceeds the folded one for the banana target. This reflects the fold-boundary residual spike entering the ρ-core at larger ρ, consistent with the diagnostic analysis in Section 5.2.

D.5 Condition verification for Theorem 4.6

We verify conditions (M′), (W), (R), and the sufficient condition (17) empirically for the Gaussian mixture ($d = 10$, $s = 2$) and label-switching ($m = 4$, $p = 2$, $s = 24$) experiments.

Quantity	Mixture d=10	Label m=4
$\min_j \Delta_j$ (from QC osc gap)	≥ 4.69	≥ 5.35
$2r_1$ (folded within-mode osc)	≤ 0.32	≤ 1.58
r_F (quotient proposal residual)	≈ 0	≈ 0
$\Delta = \min_j \Delta_j - 2r_1 - r_F$	≥ 4.37	≥ 3.77
$2(M_F\varepsilon_F^* - M_U\varepsilon_U^*)^+$	≈ 0	≈ 0
Sufficient condition (17) satisfied?	Yes	Yes
Δ/C_U	0.90	0.87

In both cases, the cross-mode deficiency Δ_j is much larger than the folded within-mode oscillation r_1, and the quotient proposal residual r_F is negligible. The sufficient condition (17) is easily satisfied, confirming that the diagnostic improvement in Theorem 4.6 holds.

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